## BEFORE

**No one knows when human beings first appeared. Our only clues lie in fossils and stone tools. The journey started some time around six million years ago (mya) in Africa.**

**THE HUMAN FAMILY**
Humans are classified as primates, a group that includes apes and monkeys. **Our closest living relatives are chimpanzees**, with whom we share almost 99 percent of our genes, but this tiny genetic difference is what makes us so far removed from apes.

CHIMPANZEE

**OUR ROOTS**
***Sahelanthropus tchadensis* 18 ››**, found at the southern edge of the Sahara in Chad, and dating to between 6 and 7 mya, may be the earliest human ancestor. Although very early, this skull seems more advanced in some ways than later species and it is **unclear how it fits into the evolutionary story**. Other very early ancestors about whom very little is known include ***Orrorin tugenensis*** and ***Ardipithecus ramidus***. Some of these species came to a **dead end on the human family tree**. Others may have led directly to our own ancestors.

**THE MOLECULAR CLOCK**
Evolutionary biologists have developed a way of dating the evolution of more than 60 primate species. It is known as the molecular clock. The clock starts with the last common ancestor of all primates about 63 mya, and dates the **split between chimpanzees and humans to about 6.2 mya**. This is the moment when the human story truly begins.

The "cradle of humankind"
Olduvai Gorge in Tanzania is the most important prehistoric site in the world, where many finds that have furthered our knowledge of early human evolution have been made. The oldest artifacts found at the gorge–stone flakes and tools–are 2 million years old.

“**Human consciousness arose** but a **minute before midnight** on the geological clock.”

STEPHEN JAY GOULD, EVOLUTIONARY BIOLOGIST, 1992

# Our Remote Ancestors

**The evolution of modern humans extends back millions of years. It is not easy to trace, as our evidence comes from scattered, unrelated finds, making it difficult to form a cohesive picture. The dominance of *Homo sapiens* is a comparatively recent development.**

In the 19th century, Charles Darwin, the father of the theory of evolution by natural selection (see pp.340–41), identified tropical Africa as the cradle of humankind. Pioneering paleontologists Louis and Mary Leakey found evidence of this in the 1950s with discoveries in Olduvai Gorge, a deep gash in the eastern Serengeti Plains in Tanzania, East Africa (see left). It was in East Africa that our human ancestors evolved at least 4.5 mya (million years ago). A wide range of fossil finds provide evidence of a remarkable diversity of early hominins that flourished in this area.

**HOMININ The term used to refer to all early humans and their ancestors, including *Homo erectus*, *Homo ergaster*, *Homo neanderthalis* and *Homo sapiens*. Also includes all the Australopithecines, *Paranthropus boisei*, and *Ardipithecus*.**

### Earliest ancestors

One of the earliest known human ancestors is a small forest-living primate named *Ardipithecus anamensis*, which flourished in Afar, Ethiopia, some 4.5 mya. *Ardipithecus* was probably the ancestor of the Australopithecines—highly diverse hominins that appeared for the first time one million years later. The earliest found, *Australopithecus afarensis*, was famously nicknamed "Lucy" by the archaeologists who found her in 1974. Although it seems that this long-limbed hominin spent a great deal of time in the trees, some well-preserved footprints reveal that the species was bipedal (walked on two feet) (see p.18). As such, "Lucy" is an important link between us and our tree-dwelling ancestors.

### The next generation

By 3 mya, the Australopithecines had diversified into many forms. They flourished throughout much of sub-Saharan Africa, especially in more open grasslands. These early humans were fully bipedal. Nimble and fleet of foot, species including *Australopithecus africanus* were skilled at scavenging meat from predator kills. Their brain size was also larger than their predecessors'.

### The first humans

Ancestors of modern humans appeared about 2 mya in eastern Africa, quickly spreading to the west. Tools dating from 1.8 mya have been found in a dry stream bed at Koobi Fora on the shore of Lake Turkana, Kenya. The tools were made of stone from several miles away. It is not known who the tool users were, but they may have been some of the earliest humans, possibly a group who paused here and butchered antelope.

**PALEOLITHIC A period covering the time from the first use of stone tools about 2.5 mya to the beginning of agriculture in about 10,000 BCE.**

### Handyman

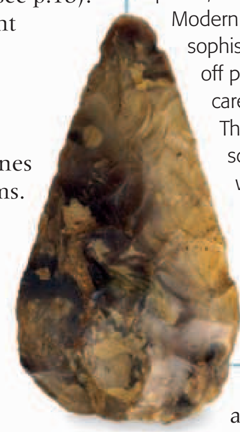
Clearer evidence of the earliest toolmakers and their descendants has been found on the ancient lake beds at Olduvai Gorge in Tanzania. The tools are thought to date from about 1.8 mya, and were made by *Homo habilis* ("handy man"), who left what could be the remains of a camp by a lake, including a scatter of stone tools and broken animal bones. *Homo habilis* probably slept in trees, in relative safety from lions and other dangerous animals. In this predator-rich environment, humans were both the hunters and the hunted. The evidence from the Olduvai camp suggests that *Homo habilis* was breaking up parts of animal carcasses scavenged from predator kills.

At about the same time, what could be termed the first true human had appeared. Large-brained, with a receding forehead, and prominent brow ridges *Homo ergaster* had strong limbs similar to those of modern humans. These newcomers were hunters rather than scavengers. ››

INVENTION

## STONE TOOLS

*Homo habilis* used the simplest stone technology, which was refined by *Homo erectus* into stone axes and cleaving tools for particular tasks such as butchering animals. The Neanderthals were the first to mount scrapers, spear-points, and knives in wooden handles. Modern humans developed more sophisticated technology, punching off parallel-sided blanks from carefully prepared flint nodules. They turned these blades into scrapers, chisels, and borers to work antlers, bone, and leather. After the Ice Age (see pp.22–23), hunters added tiny stone barbs to their arrows.



FLINT HAND-AX

DISCOVERY

## FIRE

Fire is one of the most important discoveries ever made. Possibly around 1.8 million years ago and certainly by 500,000 years ago—the date is uncertain—early humans tamed fire, perhaps by taking branches from a blazing tree caused by a lightning strike. Creating fire at will was another step forward. The control of fire enabled humans to live in cold environments, and in deep caves, and provided protection against predators. The use of fire to cook also led to a greater variety of foods in the diet.